

HRMS – Company Holiday

Technical & Developer Documentation

Status: Coding complete; functional and edge-case testing pending.

Purpose: Detailed reference for developers, architects, testers and HR users covering CompanyHoliday management, APIs, workflow and leave-day calculation.

1. Business Purpose

A company can define mandatory holidays for a specific Leave Year. When an employee requests leave, eligible company holidays inside the requested date range are excluded from the calculated leave days.

The model is Company → LeaveYear → CompanyHoliday. LeaveYear already contains CompanyId, so CompanyHoliday does not need a duplicate CompanyId for the current design.

2. Data Model

LeaveYear: Id, CompanyId, Code, Name, StartDate, EndDate, StatusId, audit fields, IsDeleted.

CompanyHoliday: Id, LeaveYearId, HolidayDate, Name, HolidayType, IsOptional, IsActive, audit fields, IsDeleted.

Database rule: unique index on (LeaveYearId, HolidayDate), so one date can occur only once within a LeaveYear.

3. Tested Leave Year

FY2027: LeaveYearId = 8BB8836A-3920-4DA2-41CF-08DEFD305CC8; CompanyId = 023FF1F8-4DD0-4942-1955-08DEFD16782B; period = 2027-04-01 to 2028-03-31.

4. Company Holiday CRUD APIs

Create: POST /api/company-holidays

```
{ "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8", "holidayDate": "2027-04-30", "name": "Company Holiday Update Test", "holidayType": "COMPANY", "isOptional": false }
```

Tested created ID: FFC4F188-EAC3-4B38-0AB3-08DEFF4B3B2E.

List: GET /api/company-holidays

```
{ "items": [{ "id": "58400962-69fe-4253-bba4-08defed654d1", "leaveYearId": "8BB8836A-3920-4DA2-41CF-08DEFD305CC8", "holidayDate": "2027-04-23", "name": "Company Holiday Test", "holidayType": "COMPANY", "isOptional": false, "isActive": true }], "totalRecords": 1, "pageNumber": 1, "pageSize": 10, "totalPages": 1 }
```

Update: PUT /api/company-holidays/{id}. **Delete:** DELETE /api/company-holidays/{id}. Exact controller routes should be verified against the project's Swagger/controller implementation before external publication.

5. Holiday Eligibility

The current calculation considers a holiday only when: same LeaveYearId; HolidayDate is within FromDate/ToDate; IsActive = true; IsOptional = false; IsDeleted = false.

Optional holidays are therefore not automatically deducted by the current calculation.

6. Leave Calculation Architecture

LeaveRequestBusinessRules keeps the existing CalculateTotalDays() as the baseline.

CalculateTotalDaysAsync() adds data-driven CompanyHoliday exclusion. Create, Submit and Update all use the new asynchronous calculation.

Flow: API → Command → Business Rules → CalculateTotalDaysAsync → Find eligible CompanyHolidays → calculate final TotalDays → persist LeaveRequest.

7. Create Workflow

Validate employee → validate LeaveYear → validate LeaveType → validate day parts → validate dates/day parts → check overlap → CalculateTotalDaysAsync → create DRAFT LeaveRequest.

Create endpoint: POST /api/leave-requests. The exact request contract should follow the existing CreateLeaveRequestCommand/controller.

8. Submit Workflow

Submit revalidates the request and recalculates TotalDays before moving DRAFT → PENDING. This protects against a holiday calendar changing after the request was created.

Example: POST /api/leave-requests/DAAA731C-86A6-4F2F-1EB4-08DEFF48E482/submit. The command contains only Id, so no business JSON body is required; an empty {} may be accepted depending on controller binding.

9. Update Workflow

Update recalculates the new date range using CalculateTotalDaysAsync. Existing status and overlap rules remain unchanged.

```
{ "id": "A6562C90-4F07-49D9-B1AF-08DEFED676EE", "leaveTypeId": "9A5FF5C9-49DF-43F4-5107-08DEFD337F69",  
  "startDayPartId": "30000000-0000-0000-0000-000000000001", "endDayPartId":  
  "30000000-0000-0000-0000-000000000001", "fromDate": "2027-04-28", "toDate": "2027-05-02", "reason":  
  "Company holiday update test" }
```

10. Calculation Example

Leave: 2027-04-28 → 2027-05-02 28 = 1.00 29 = 1.00 30 = 0.00 ← Company Holiday 01 = 1.00 02 = 1.00
Final TotalDays = 4.00

The tested Create scenario used 2027-04-23 as the holiday for 2027-04-21 through 2027-04-25 and produced 4.00. The tested Update scenario used 2027-04-30 and produced 4.00 for 2027-04-28 through 2027-05-02.

11. SQL Verification

```
SELECT Id, LeaveYearId, FromDate, ToDate, TotalDays, StatusId, Reason FROM LeaveRequests WHERE Id =  
'A6562C90-4F07-49D9-B1AF-08DEFED676EE';
```

```
SELECT Id, LeaveYearId, HolidayDate, Name, HolidayType, IsOptional, IsActive, IsDeleted FROM  
CompanyHolidays WHERE LeaveYearId = '8BB8836A-3920-4DA2-41CF-08DEFD305CC8' ORDER BY HolidayDate;
```

12. Tested Records

Holiday 2027-04-23: 58400962-69FE-4253-BBA4-08DEFED654D1. Holiday 2027-04-30:

FFC4F188-EAC3-4B38-0AB3-08DEFF4B3B2E. Create/Submit test request:

DAAA731C-86A6-4F2F-1EB4-08DEFF48E482, result 4.00 and submitted to PENDING. Update test request:
A6562C90-4F07-49D9-B1AF-08DEFED676EE, result 4.00.

13. Existing Workflow Protections

Example 409: "Only draft leave requests can be updated." This is an existing status rule, not a holiday-calculation failure. Example 409: "Employee already has a pending or approved leave request for the selected dates." This is the existing overlap rule.

14. Architecture / Design Rationale

CompanyHoliday is scoped through LeaveYear, avoiding a duplicate company relationship. One reusable business-rule method is used by Create, Submit and Update, avoiding three different calculations. Existing day-part calculation remains the baseline.

15. Testing Still Required

Coding is complete. Before production sign-off test: holiday on first day; holiday on last day; single-day holiday; half-day + holiday; multiple holidays; optional holiday; inactive holiday; soft-deleted holiday; holiday from another LeaveYear; weekends/non-working days; and authorization/company isolation.

16. Production Checklist

- Complete edge-case tests
- Verify exact controller routes/Swagger
- Verify authorization and company isolation
- Verify audit fields
- Add unit tests for calculation
- Add integration tests for Create/Submit/Update
- Review migration/schema changes
- Commit after tests pass

17. Developer Quick Reference

Eligible holiday = same LeaveYear + inside requested range + active + mandatory (not optional) + not deleted. Create, Submit and Update call CalculateTotalDaysAsync. Main proven scenario: five calendar days with one middle mandatory company holiday = 4.00 leave days.